INCH-POUND
MS45904C
w/AMENDMENT 1
15 September 2026
SUPERSEDING
MS45904C
07 July 2022

DETAIL SPECIFICATION SHEET

WASHER, LOCK, INTERNAL AND EXTERNAL TOOTH

The requirements for acquiring the described herein shall consist of this specification sheet and specification ASME B18.21.1.

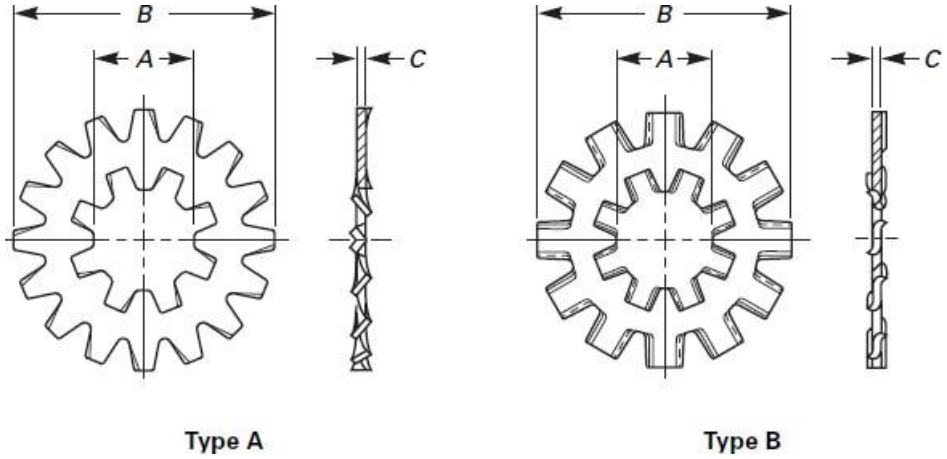


FIGURE 1. Acceptable types of internal and external tooth lock washer.

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REQUIREMENTS:

1 Steel composition. Unless otherwise specified, washers shall be made from carbon steel, SAE 1050 through 1065 (UNS G10500–G10650), (see SAE J403, Table 1).

2 Hardness. Washers shall be hardened to HRC 40 through HRC 50 or equivalent measuring method.

3 Internal and external teeth. Internal and external teeth shall be either Type A or Type B in accordance with ASME B18.21.1.

4 Protective finish. Zinc-Nickel plate in accordance with MIL-PRF-32647, Type K20, or Cadmium[1/] in accordance SAE AMS-QQ-P-416, Type II, Class 2.

[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647 is the preferred method.

5 Conformance. Certificates of compliance to paragraph 1, 2, 3 and 4 shall be provided to verify conformance.

6 Part or Identifying Number (PIN). The PIN consists of the following:

MS45904 - 818

- 818: Dash number from Table I indicating size/dimensions (cross-referenced to Figure 1.)
- MS45904: Specification sheet number

Example: MS45904-818 is a nominal .250 inch internal and external tooth lock washer, outside diameter (OD) approximately .750 inch with Zinc-Nickel plating.

TABLE I. Dash numbers and dimensions (inch).

DASH NUMBER (Zn-Ni)	NOMINAL SIZE	INSIDE DIAMETER A		OUTSIDE DIAMETER B		THICKNESS C	
		MAX	MIN	MAX	MIN	MAX	MIN
801	.112	.123	.115	.475	.460	.021	.016
804	.138	.150	.141	.510	.495	.028	.023
807	.164	.176	.168	.610	.580	.034	.028
808	.164	.176	.168	.690	.670	.034	.028
809	.164	.176	.168	.760	.740	.034	.028
810	.190	.204	.195	.610	.580	.034	.028
811	.190	.204	.195	.690	.670	.040	.032
812	.190	.204	.195	.760	.740	.040	.032
813	.190	.204	.195	.900	.880	.040	.032
818	.250	.267	.256	.760	.725	.040	.032
819	.250	.267	.256	.900	.880	.040	.032
820	.250	.267	.256	.985	.965	.045	.037
821	.250	.267	.256	1.070	1.045	.045	.037
822	.3125	.332	.320	.900	.865	.040	.032
823	.3125	.332	.320	.985	.965	.045	.037
824	.3125	.332	.320	1.070	1.045	.050	.042
826	.375	.398	.384	.985	.965	.045	.037
827	.375	.398	.384	1.070	1.045	.050	.042
828	.375	.398	.384	1.155	1.130	.050	.042
829	.375	.398	.384	1.260	1.220	.050	.042
834	.500	.530	.512	1.260	1.220	.055	.047
836	.500	.530	.512	1.410	1.380	.060	.052
837	.500	.530	.512	1.620	1.590	.067	.059
842	.625	.663	.640	1.410	1.380	.060	.052
843	.625	.663	.640	1.620	1.590	.067	.059

NOTE: Washers cannot be used when directly converted to metric; as such, no metric conversion is given.

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TABLE II. Inactive and replacement part numbers.

PIN			
Inactive for new Design	**Zinc-Nickel Replacement**	**Inactive for new Design**	**Zinc-Nickel Replacement**
MS45904-	MS45904-	MS45904-	MS45904-
51	801	74	824
52	*	75	*
53	*	76	826
54	804	77	827
55	*	78	828
56	*	79	829
57	807	80	*
58	808	81	*
59	809	82	*
60	810	83	*
61	811	84	834
62	812	85	*
63	813	86	836
64	*	87	837
65	*	88	*
66	*	89	*
67	*	90	*
68	818	91	*
69	819	92	842
70	820	93	843
71	821	94	*
72	822	95	*
73	823		

* Denotes no replacement part number, previously designated as inactive for new design.

7 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids. Copies of MIL-PRF-32647 are available online at https://assist.dla.mil. Copies of ASME B18.21.1 are available online at https://www.asme.org. Copies of SAE J403 and SAE AMS-QQ-P-416 are available online at https://www.sae.org.

8 Design. For design feature purposes, this document takes precedence over all other documents.

9 Dimensions. Unless otherwise specified, all dimensions are in inches.

10 Amendment notations. The margins of this document are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

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Army – AT

(Project 5310-2026-012)

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